INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

END-SEMESTER EXAMINATION 2021-22

Academic Session: 2021-2022

Programme: B.Tech. (Computer Science & Engineering) | **Semester: Semester III**

Course Code: BCS-301 | **Course Title: Data Structures**

Time Allowed: 3 Hours | Maximum Marks: 50

Note: Attempt all sections as per instructions. Assume missing data suitably, if any.

SECTION A (Attempt All Questions - 2 Marks each)

Q1(a). Explain Asymptotic notations: Big-Oh (O), Omega, and Theta.

Q1(b). What is a sparse matrix? How is it efficiently represented in memory?

Q1(c). Write recursive relation for Towers of Hanoi problem.

Q1(d). Define Complete Binary Tree vs Strictly Binary Tree.

Q1(e). What is the difference between Array and Linked List memory allocation?

SECTION B (Attempt Any Three - 5 Marks each)

Q2. Explain Heap Sort algorithm and illustrate with array: [45, 12, 85, 32, 89, 39, 69, 44].

Q3. Describe Dequeue and Priority Queue with standard operations and applications.

Q4. Explain Binary Search Tree insertion and deletion algorithms.

SECTION C (Attempt Any Two - 10 Marks each)

Q5. Write complete C functions to implement circular singly linked list traversal and deletion.

Q6. Discuss Dijkstra’s algorithm for finding Single Source Shortest Path with an example.

Q7. Describe B-Tree of order m. Explain insertion and node-splitting mechanism.

========== END OF QUESTION PAPER ==========